

Regular consumption of pulses for 8 weeks reduces metabolic syndrome risk factors in overweight and obese adults.



Pulses are low in energy density, supporting their inclusion in the diet for the management of risk factors of the metabolic syndrome (MetSyn). The aim of the present study was to describe the effects of frequent consumption (five cups/week over 8 weeks) of pulses (yellow peas, chickpeas, navy beans and lentils), compared with counselling to reduce energy intake by 2093 kJ/d (500 kcal/d), on risk factors of the MetSyn in two groups (nineteen and twenty-one subjects, respectively) of overweight or obese (mean BMI 32.8 kg/m²) adults. Body weight, waist circumference, blood pressure, fasting blood parameters and 24 h food intakes were measured at weeks 1, 4 and 8. Blood glucose, insulin, C-peptide, glucagon-like peptide-1 (GLP-1) and ghrelin were measured after a 75 g oral glucose load at weeks 1 and 8. At week 8, both groups reported reductions in energy intake, waist circumference, systolic blood pressure, glycosylated Hb (HbA1c) and glucose AUC and homeostasis model of insulin resistance (HOMA-IR) following the glucose load ($P < 0.05$). However, HDL, fasting C-peptide and insulin AUC responses were dependent on diet ($P < 0.05$). HDL and C-peptide increased by 4.5 and 12.3 %, respectively, in the pulse group, but decreased by 0.8 and 7.6 %, respectively, in the energy-restricted group. Insulin AUC decreased in both females and males on the energy-restricted diet by 24.2 and 4.8 %, respectively, but on the pulse diet it decreased by 13.9 % in females and increased by 27.3 % in males ($P < 0.05$). In conclusion, frequent consumption of pulses in an ad libitum diet reduced risk factors of the MetSyn and these effects were equivalent, and in some instances stronger, than counselling for dietary energy reduction.